

Certified lower bounds on the complementary gap from the state of a matching decoder

Sumeet Shirkure

Department of Electrical and Computer Engineering, University of Central Florida

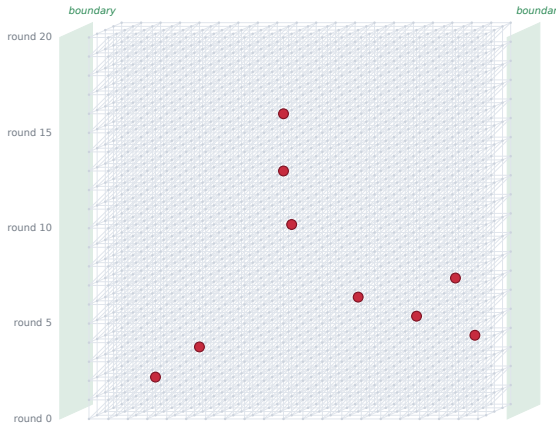
September 2026

Soft output for surface-code memory: what the decoder already knows after it has matched,
and how to turn it into a rigorous number without walking the empty spacetime volume.

Surface-code memory decoding is a matching problem

(a) Fired detectors inside G

One decoding graph of the rotated surface code under circuit-level noise: space-like, time-like and diagonal (hook) edges over $d \times d \times N$



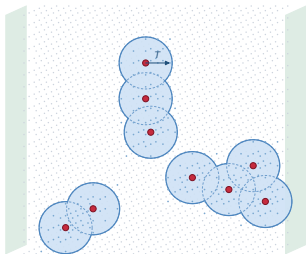
the shot is n defects; the empty volume between them carries capacitance information but requires $O(r^3)$ CPU cycles to certify a capacitance of $O(r)$

- Detector graph $G = (V, E, w)$: one node per detector in a $d \times d \times N$ spacetime volume; edges are the error mechanisms of the DEM, weighted $w_e = \ln \frac{1-p_e}{p_e}$.
- A shot fires a set of **defects** $D \subseteq V$. Every error mechanism flips ≤ 2 detectors, so a correction is a **perfect matching** of D into pairs or into the boundaries L, R .
- Minimum weight $\Leftrightarrow$ most likely error: the MWPM decoder (sparse blossom / *pymatching*).
- The two boundaries L, R define the logical: a correction touching L an odd number of times is a different logical class.

The typical syndrome is a handful of small, well separated clusters

(b) One ball of radius T per fired detector

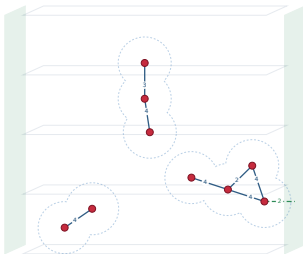
$B_T(u) = \{v : d_G(u, v) \leq T\}$, read from ball tables compiled once per DEM
(tables are stored out to $R \leq 2T$, so a lookup from either endpoint is $O(1)$)
their union is a thin tube around the syndrome



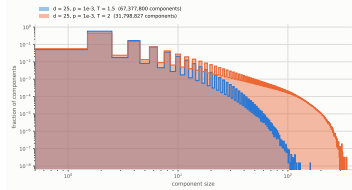
two balls meet $\Leftrightarrow d_G(u, v) \leq 2T \Leftrightarrow$ the two defects' duals can still interact at horizon T

(c) The defect manifold graph $\mathcal{H}$

nodes = this shot's defects; edge (u, v) iff $d_G(u, v) \leq 2T$, of weight $w_{uv} = d_G(u, v)$ — an induced subgraph of the metric



truncated sparse blossom runs here at horizon T and commits what it has paired



Component sizes of $\mathcal{H}$ across 10^6 shots, $d = 25$,
 $p = 10^{-3}$.

- At $p = 10^{-3}$ a defect has a partner within $\text{dist} \leq 2T$ ($T \approx 2$ edges) almost always; the union of radius- T balls is a thin tube around the syndrome.
- Defect graph $\mathcal{H}$: edge (u, v) iff $\text{dist}_G(u, v) \leq 2T$. It splits into components of size 2, 4, $\dots$; a T -truncated blossom matches each component independently (*SpecMatching*).
- **Consequence:** $> 99\%$ of the spacetime volume is empty. The decoder never visits it, and finishes in $\approx 8 \mu\text{s}$ at $d = 17$ and $\approx 24 \mu\text{s}$ at $d = 31$ on a few cores.

Soft output: the complementary gap

Definition

Let M^* be the minimum-weight matching, of weight W^* , and M^c the minimum-weight matching of the *other* logical class, of weight W^c . The complementary gap is

$$\Delta = W^c - W^* \geq 0.$$

Why it is useful

- Weights are log-likelihoods, so $\Delta = \ln \frac{\Pr(M^* \mid \text{syndrome})}{\Pr(M^c \mid \text{syndrome})}$: a per-shot confidence in the correction.
- Discard shots with $\Delta < \tau$: post-selection for magic-state cultivation, hierarchical codes, decoder switching.
- Empirically $\Pr(\text{logical error} \mid \Delta) \approx e^{-\Delta}$ over many orders of magnitude.

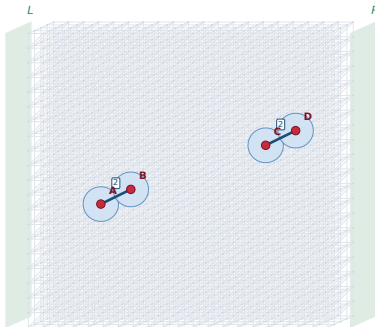
How it is usually computed

- Decode once, then decode *again* with the logical class forced (e.g. add a defect on the boundary L).
- The second decode grows a slab from the whole boundary sheet across the patch: its cost scales with d and with Δ itself.
- Doubles (or worse) the decode latency, exactly on the high-gap shots that the application keeps.

A worked example: four defects, two tight pairs

(a) Minimum-weight matching M^*

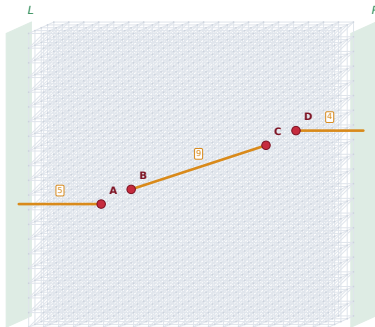
$W^* = 4$: dual radii y_v shown as balls, matched edges are tight:
 $\text{dist}(u, v) = y_u + y_v$



class of M^* : even number of edges touching $L \Rightarrow$ no logical flip

(b) Best matching of the other class, M^c

$W^c = 18$: one edge touches L (odd), so this correction differs from M^* by a logical operator



complementary gap $\Delta = W^c - W^* = 18 - 4 = 14$

- Unit-weight edges (space, time and one hook family); $\text{dist}(L, \cdot)$, $\text{dist}(\cdot, R)$ are the boundary distances.
- $M^* = \{AB, CD\}$, $W^* = 4$. Best odd-class matching $M^c = \{LA, BC, DR\}$, $W^c = 18$. So $\Delta = 14$: this shot is e^{14} times more likely to be in the trivial class.
- Sparse blossom also returns **dual variables** y_v , drawn as balls of radius y_v (here $y_v = 1$ for every defect). Next slide: what they are and what they certify.

Background: the matching LP, dual charge and weak duality

Primal. Choose edges so every defect is covered exactly once: pairs (u, v) , or (u, L) , (u, R) . Minimise $W(M) = \sum_{e \in M} \text{dist}(e)$.

Dual. A number $z_S \geq 0$ on some sets S of nodes. Singletons carry the ball radii, $z_{\{v\}} = y_v$ (L, R get none).

Dual charge

Edge e crosses set S if exactly one endpoint of e is in S .

$$\text{charge}(e) = \sum_{S: e \text{ crosses } S} z_S.$$

With singletons only: $\text{charge}(u, v) = y_u + y_v$,
 $\text{charge}(u, L) = y_u$.

Feasible: $\text{charge}(e) \leq \text{dist}(e)$ for every possible edge e .

Reduced cost: $\sigma(e) = \text{dist}(e) - \text{charge}(e) \geq 0$.

Weak duality

Let $c_M(S)$ = number of edges of M crossing S . If every M in some family has $c_M(S) \geq 1$ for all S , then

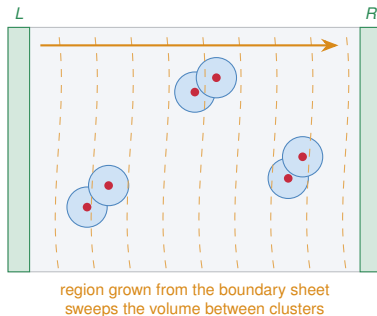
$$W(M) \geq \sum_{e \in M} \text{charge}(e) = \sum_S z_S c_M(S) \geq \sum_S z_S.$$

The first step is feasibility, the last is $c_M(S) \geq 1$.

- Every perfect matching crosses each $\{v\}$ once, so $W^* \geq \sum_v y_v$. Sparse blossom returns y with equality, so every M^* edge is **tight**: $\sigma = 0$. (Example: $\sum_v y_v = 4 = W^*$.)
- **Plan:** use exactly two facts from the decoder's final state, tightness of M^* edges and $\sigma \geq 0$ everywhere, to turn Δ into a non-negative shortest-path problem.

The catch: the second decode sweeps the empty volume

- Forcing the other class means adding a defect on the boundary sheet (or on a detector attached to the logical observable) and decoding again.
- Sparse blossom then grows a region from that sheet. Its front is the whole sheet, area $\sim d \times N$ detectors, and it must keep growing until the extra dual it has accumulated reaches Δ .
- On the high-gap shots an application keeps, most of that growth happens in the empty space between clusters: exactly the $\gtrsim 99\%$ of the volume the first decode never touched.
- So the gap typically costs more than the decode it annotates, and the cost grows with d even though the information we want sits near the defects.

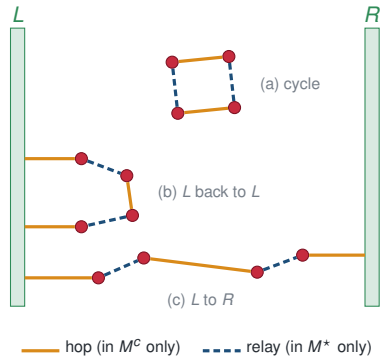


What we want instead

Compute Δ , or a certified lower bound on it, from the *first* decode's final state, pricing the empty space between clusters without visiting it.

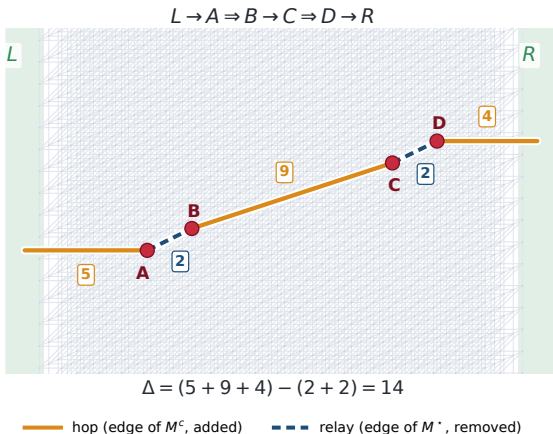
Step 1: the symmetric difference $M^* \oplus M^c$

- $M^* \oplus M^c$: edges in exactly one of the two matchings.
Hops: edges of $M^c \setminus M^*$ (added). **Relays**: edges of $M^* \setminus M^c$ (removed).
- Each defect has one M^* edge and one M^c edge, so its degree in $M^* \oplus M^c$ is 0 (same edge) or 2 (one hop, one relay). L , R may have any degree.
- Hence $M^* \oplus M^c$ splits into **components** (maximal connected pieces), each alternating hop, relay, hop, ..., of three kinds: (a) cycles, (b) paths from a boundary back to itself, (c) L – R paths.
- Flipping one component (swap its hops and relays in M^*) gives another valid matching, heavier by $\sum_{\text{hops}} \text{dist} - \sum_{\text{relays}} \text{dist} \geq 0$.
- Only (c) changes the parity of $\deg(L)$: the class flips iff the number of L – R paths is odd. Any other component can be dropped without changing the class or raising the weight. So for the optimal M^c , $M^* \oplus M^c$ is a **single L – R path**.



A general $M^* \oplus M^c$ could contain all three kinds.
For the *optimal* M^c only one piece of kind (c) survives.

The gap is a shortest alternating path



In the example $M^* \oplus M^c$ is the one path $L \rightarrow A \Rightarrow B \rightarrow C \Rightarrow D \rightarrow R$ ($\rightarrow$ hop, $\Rightarrow$ relay).

In general, by the previous slide,

$$\Delta = \min_{P \text{ simple alternating } L \rightarrow R} \left[\sum_{\text{hops}} \text{dist} - \sum_{\text{relays}} \text{dist} \right],$$

where *alternating* means hops and relays alternate, and the relays are exactly the M^* edges it passes through. The end edges touch L or R and may be of either kind (a defect matched to L by M^* can be released by a relay).

A shortest-path problem, but relays carry *negative* weight and the path must alternate, so not yet Dijkstra. Next: reduced costs remove both obstacles.

Step 2: reduced costs make every hop non-negative

For now only vertex duals y_v (no blossoms). Alternating path $P = L \rightarrow v_1 \Rightarrow v_2 \rightarrow v_3 \Rightarrow v_4 \rightarrow \dots \rightarrow R$. Relays are M^* edges, hence tight: $\text{dist}(v_1, v_2) = y_{v_1} + y_{v_2}$.

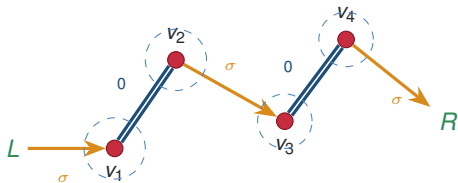
$$\text{cost}(P) = \sum_{\text{hops}} \text{dist}(u, v) - \sum_{\text{relays}} (y_u + y_v).$$

Each interior defect is on one hop and one relay: move its y from the relay onto the hop, and the sum **telescopes**:

Reduced cost

$$\text{cost}(P) = \sum_{\text{hops}} \underbrace{(\text{dist}(u, v) - y_u - y_v)}_{\sigma(u, v) \geq 0}, \quad \sigma(L, v) = h_L(v) - y_v.$$

$\sigma \geq 0$ by dual feasibility, and each relay's cost has been absorbed into its two neighbouring hops. The decoder's own duals supply this reweighting for free.



After reweighting every hop costs its reduced cost $\sigma \geq 0$ (ball to ball) and every relay costs 0: the dashed radii cancel pairwise.

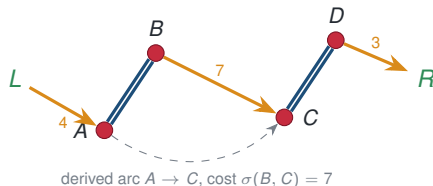
One Dijkstra on the defect graph: land on v , leave from $\text{mate}(v)$

Derived directed graph

Nodes: the defects, plus L, R . Arc $v \rightarrow u$ with cost $\sigma(\text{mate}(v), u)$; arcs $L \rightarrow u$ with cost $\sigma(L, u)$; arcs $v \rightarrow R$ with cost $\sigma(\text{mate}(v), R)$. Landing on v with $\text{mate}(v) = R$ ends the path; a defect with $\text{mate} = L$ is a free starting point.

$\Delta_\sigma =$ shortest $L \rightarrow R$ path in this graph.

- Non-negative costs $\Rightarrow$ plain Dijkstra on $|D| + 2$ nodes. The arc matrix is the $|D| \times |D|$ matrix of reduced costs with its rows permuted by the matching.
- Every simple alternating path is a walk in the derived graph with equal cost, so $\Delta_\sigma \leq \Delta$ (blossom-free here; blossoms next). Extra walks that reuse a relay only lower the minimum.



landing on A prices the next hop from $B = \text{mate}(A)$;
 $\Delta_\sigma = 4 + 7 + 3 = 14 = \Delta$ in the example

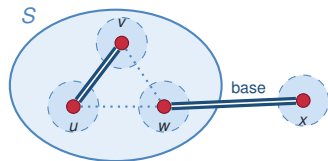
What we have so far

$$\Delta_\sigma \leq \Delta,$$

computed from the first decode's final state by one Dijkstra on the defects. The remaining obstacle: σ needs $\text{dist}(u, v)$ for far-apart pairs.

Blossoms (1/3): total radius and shared mass

- Sparse blossom also returns $y_S \geq 0$ on **blossoms** S : odd sets of defects, nested or disjoint. Every perfect matching crosses an odd set, so it is a legitimate dual set.
- **Total radius** $\rho_v = y_v + \sum_{S \ni v} y_S$: how far v 's region reaches.
- **Shared mass** $\beta(u, v) = \sum_{S \ni u, v} y_S$: the dual on blossoms that contain *both* u and v (0 if they share none).
- Charge on $(u, v) = y_u + y_v +$ blossoms with *exactly one* endpoint inside. $\rho_u + \rho_v$ counts each blossom containing both endpoints **twice**, yet (u, v) lies inside it and does not cross it. So



edge	crosses	charge
(u, v)	$\{u\}, \{v\}$	$y_u + y_v$
(w, x)	$\{w\}, \{x\}, S$	$y_w + y_S + y_x$
$\rho_u + \rho_v = y_u + y_v + 2y_S, \quad \beta(u, v) = y_S$		

Blossom $S = \{u, v, w\}$: dashed balls have radius y_v ; the shaded band around them has width y_S . Dotted edges are tight edges inside the blossom.

Charge and reduced cost with blossoms

$$\text{charge}(u, v) = \rho_u + \rho_v - 2\beta(u, v),$$

$$\sigma(u, v) = \text{dist}(u, v) - \rho_u - \rho_v + 2\beta(u, v) \geq 0.$$

Blossoms (2/3): why the $+2\beta$ matters, and what is left after telescoping

Without the correction. Pricing hops by $g(u, v) = \text{dist} - \rho_u - \rho_v$ gives

$$g(u, v) = \sigma(u, v) - 2\beta(u, v) < 0$$

for many pairs inside a blossom: their regions overlap by the blossom dual they share. Negative hop weights break Dijkstra.

The base. Complementary slackness: if $y_S > 0$, exactly **one** edge of M^* crosses S . Call it the **base** of S . (Odd S : all but one defect is matched inside.)

Question

Is this non-negative? If yes, pricing hops by $\sigma \geq 0$ underestimates every path, and Dijkstra is safe. Next slide: yes, one blossom at a time.

Telescoping again. Relays are tight:

$\text{dist}(a, b) = \rho_a + \rho_b - 2\beta(a, b)$. The radii ρ cancel at every interior defect exactly as before, and the shared masses remain:

$$\text{cost}(P) = \sum_{\text{hops}} \sigma(h) + 2 \left[\sum_{\text{relays}} \beta(r) - \sum_{\text{hops}} \beta(h) \right].$$

An edge contributes y_S to β exactly when it lies *inside* S (both ends in S), so the bracket equals

$$\sum_S y_S (r_S(P) - h_S(P)),$$

with r_S, h_S the numbers of relays and hops of P inside S .

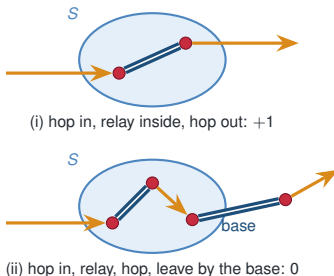
Segment

A **segment** of P in blossom S : a maximal run of consecutive vertices of P lying in S . As $L, R \notin S$, it is entered by one edge crossing S and left by another; the edges between are P 's edges inside S .

Per segment, relays inside minus hops inside:

- (i) **in and out by hops:** hops are flanked by relays, so inside runs relay, hop, ..., relay: **+1**.
- (ii) **one crossing is the base:** inside, relays and hops alternate in equal number (e.g. hop in, relay, hop, base out): **0**.
- (iii) **both crossings relays:** impossible; only the base crosses S .

With $n_S(P)$ = number of type-(i) segments of P in S :
 $r_S(P) - h_S(P) = n_S(P) \geq 0$.



Result

$$\text{cost}(P) = \sum_{\text{hops}} \sigma(h) + 2 \sum_S y_S n_S(P) \geq \sum_{\text{hops}} \sigma(h).$$

So $\Delta_\sigma \leq \Delta$ on every shot, blossoms included.
 It misses $2y_S$ only on hop-in, hop-out passes.

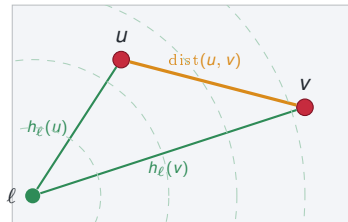
Step 3: price long hops with a certified lower bound

Landmark potentials

Pick a few landmarks ℓ (boundary sheets L , R , time faces, corners) and precompute once per DEM the potential $h_\ell(x) = \text{dist}(\ell, x)$ for every node x . By the triangle inequality, for every pair

$$\text{LB}(u, v) = \max_{\ell} |h_\ell(u) - h_\ell(v)| \leq \text{dist}(u, v).$$

- $O(1)$ per pair at run time, $O(|V|)$ memory per landmark; nothing about the DEM is assumed, so it works on cultivation graphs too.
- Short pairs (within a table radius R) are *exact* from local distance tables built once per DEM. Only long hops are priced by LB.
- Boundary legs are exact: h_L and h_R are landmarks themselves.



Triangle inequality: $\text{dist}(u, v) \geq |h_\ell(v) - h_\ell(u)|$.
Dashed: level sets of h_ℓ . Each landmark certifies a direction: on an L_1 -like metric the faces make LB exact along the axes, corners tighten the diagonals.

Step 4: one dense, parallel Dijkstra from the decoder's final state

Inputs: defects D , mates, radii ρ_v , shared masses β , local distance tables, potentials h_ℓ .

1. Price every defect pair:
 $\hat{\sigma}(u, v) = \max(\widehat{\text{dist}}(u, v) - \rho_u - \rho_v + 2\beta(u, v), 0)$, with $\widehat{\text{dist}} = \text{dist}$ from a table if within R , else LB.
2. Boundary prices $\hat{\sigma}(L, u) = \max(h_L(u) - \rho_u, 0)$, likewise for R (exact).
3. Arc $v \rightarrow u$ costs $\hat{\sigma}(\text{mate}(v), u)$: rows permuted by the matching.
4. One Dijkstra on $|D| + 2$ nodes returns $\Delta_{\sigma, H}$.

Guarantee

$\widehat{\text{dist}} \leq \text{dist}$ on every pair, so $\hat{\sigma} \leq \sigma$ and

$$\Delta_{\sigma, H} \leq \Delta_{\sigma} \leq \Delta.$$

Thresholding on it never keeps a shot the exact gap would discard.

Why this is fast

- Nothing outside the defects is touched: $O(|D|^2)$ table lookups plus an $O(|D|^2)$ dense Dijkstra, independent of the volume $|V|$.
- The $|D| \times |D|$ price matrix is embarrassingly parallel: each core owns a block of rows, no synchronisation until the Dijkstra.
- With a threshold τ the Dijkstra stops early: post-selection only needs to know whether $\Delta_{\sigma, H} \geq \tau$.

Measured in *fastgap* ($d = 17, p = 10^{-3}$)

Gap stage $\approx 12 \mu\text{s}$ median on 4 threads, vs. $\approx 536 \mu\text{s}$ for the extra class-forced decode: $45\times$. Details two slides ahead.

What the landmark relaxation costs, and an upper bound for free

$$0 \leq \Delta_\sigma - \Delta_{\sigma,H} \leq \sum_{h \in P_H} (\text{dist}(h) - \widehat{\text{dist}}(h))$$

over the hops of the walk P_H the relaxed search returned.

- Only hops longer than the table radius contribute. A shot whose best walk uses only short hops loses nothing.
- Selection bias: the search minimises over lower bounds, so it prefers the hops it under-prices most. The underestimate is governed by the worst-certified direction, not the average one.
- With only the six face potentials on an L_1 lattice, LB degenerates to L_∞ : transverse motion is free. Corner landmarks fix this at a few bytes per node. Landmark placement is DEM-dependent.

An upper bound on the same walk

Let $r(u) = \rho_u - \beta(u, \text{mate}(u))$. Relays split exactly, $\text{dist}(u, \text{mate}(u)) = r(u) + r(\text{mate}(u))$, so for a *simple* alternating path P

$$\text{cost}(P) = \sum_{\text{hops}} (\text{dist}(a, b) - r(a) - r(b)) \geq \Delta.$$

Re-price only the few hops of P_H exactly (targeted shortest-path queries) to get $\Delta_h \geq \Delta$.

Every shot then carries an interval $[\Delta_{\sigma,H}, \Delta_h] \ni \Delta$:

- keep if $\Delta_{\sigma,H} \geq \tau$; discard if $\Delta_h < \tau$;
- only the “uncertain” shots in between need more work.

fastgap: measured gap latency

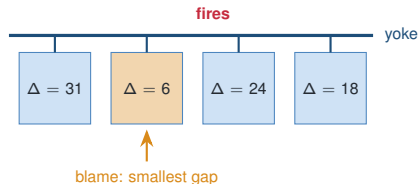
d, p	$\overline{ D }$	baseline	fastgap (μ s)		speed-up	
		(μ s)	$T=1$	$T=4$	$T=1$	$T=4$
$5, 5 \times 10^{-3}$	8.4	7.17 / 14.6	0.75 / 2.5	0.75 / 2.7	9.6 / 5.7	9.6 / 5.5
$7, 5 \times 10^{-3}$	25.3	24.7 / 44.5	2.75 / 7.1	2.62 / 5.6	9.0 / 6.3	9.4 / 7.9
$5, 10^{-3}$	1.8	3.67 / 10.0	0.12 / 0.9	0.12 / 0.9	29.3 / 10.9	29.3 / 10.9
$9, 10^{-3}$	12.0	38.5 / 62.7	1.50 / 5.9	1.54 / 5.7	25.7 / 10.7	25.0 / 11.0
$13, 10^{-3}$	38.3	158.7 / 232.8	6.71 / 14.8	5.50 / 13.0	23.7 / 15.8	28.9 / 17.9
$17, 10^{-3}$	87.8	535.9 / 808.5	18.9 / 34.2	11.9 / 21.9	28.4 / 23.6	45.0 / 37.0

Entries are median / p99, compute-only, gap stage only: the ordinary decode both methods share is excluded. Baseline = the extra class-forced PyMatching decode. T = threads. $\overline{|D|}$ = mean number of defects. Apple M5 Pro, 2×10^4 shots per row. Source: github.com/sumeetshirgure/fastgap README.

- **9–45 $\times$ faster** than the second decode, and the lead holds at p99.
- The advantage **grows with d** : the class-forced decode grows a region from the whole boundary sheet, while fastgap's work scales with $|D|^2$ only.
- **Mostly algorithmic, not parallel**: 4 threads help only at large $|D|$ ($1.6\times$ at $d = 17$).
- **Still to do**: rerun on a pinned Linux server, and measure end to end (syndrome in, verdict out).

Application: yoked surface codes

- **Yoking** (Gidney et al., *Nat. Commun.* 2025): surface-code patches in a grid, with multi-patch logical parity checks (**yokes**) on every row (1D), and also on every column (2D).
- The outer codes are $[[n, n - 2, 2]]$ (1D) and $[[n^2, n^2 - 4n + 2, 4]]$ (2D).
- **Two-level decoder:** each patch is decoded by MWPM and reports its gap Δ_i . The outer decoder is MWPM on an outer graph in which patch i is an edge of weight Δ_i .
- In 1D, when a yoke fires the outer decoder blames the patch with the **smallest gap**.
- Soft information doubles (1D) or quadruples (2D) the distance exponent; the fitted rates fall like 8^{-d} and 50^{-d} .



A row of four patches. The yoke detects that some patch failed; the gaps decide which one.

2D: each patch is an edge between its row check and its column check; the outer graph is a layered complete bipartite graph, still decoded by MWPM.

Nothing can be discarded

An underestimated gap can make an *innocent* patch look guilty. Does replacing Δ_i by $\Delta_{\sigma, H}$ break the distance doubling?

Distance preservation needs one inequality on each side

For one patch, write E for the true error, $W(E)$ for its weight, and $\mathcal{D} = \text{dist}(L, R)$ for the weight of the lightest logical. Two facts:

(F1) $W^* \leq W(E)$: E itself is a valid correction.

(F2) $W^* + W^c \geq \mathcal{D}$: opposite classes differ by an L - R chain.

Culprit (its inner decode failed): $W^c \leq W(E)$, so $\Delta \leq 2W(E) - \mathcal{D}$.

Innocent (its decode was right): $\Delta \geq \mathcal{D} - 2W^* \geq \mathcal{D} - 2W(E)$.

Conditions on the estimate $\hat{g}$ the outer decoder receives

(C) culprit side: $\hat{g} \leq \Delta$.

(I) innocent side: $\hat{g} \geq \mathcal{D} - 2W(E)$.

A lower bound gets (C) for free. It has to *earn* (I).

1D. A wrong blame of innocent B for culprit A needs $\hat{g}_B \leq \hat{g}_A$, so

$$\mathcal{D} - 2W_B \leq \hat{g}_B \leq \hat{g}_A \leq \Delta_A \leq 2W_A - \mathcal{D},$$

so $W_A + W_B \geq \mathcal{D}$. **Any matchable outer code.** Let F be the culprit set and C the set the outer MWPM flips. Then $\sum_C \hat{g} \leq \sum_F \hat{g}$, and (C), (I) give

$$\sum_{i \in C \oplus F} W_i \geq \frac{\mathcal{D}}{2} |C \oplus F| \geq \frac{\mathcal{D}}{2} D_o.$$

A failure makes $C \oplus F$ a nontrivial outer logical: $|C \oplus F| \geq D_o$.

Exponent

Failure needs fault weight $\geq \mathcal{D}$ in 1D ($D_o = 2$), $\geq 2\mathcal{D}$ in 2D ($D_o = 4$), vs. $\mathcal{D}/2$ for one patch: the same as with the exact gap.

Innocent side for $\Delta_{\sigma,H}$ (1/3): reduce to a cluster path

Goal: $\Delta_{\sigma,H} \geq \mathcal{D} - 2W^*$, which with (F1) gives (I), even with landmark-priced hops and blossoms.

Relax the hop prices. Let $\hat{g}(u, v) = \widehat{\text{dist}}(u, v) - \rho_u - \rho_v$. Since $\beta \geq 0$, every hop price satisfies $\hat{\sigma} \geq \max(\hat{g}, 0)$.

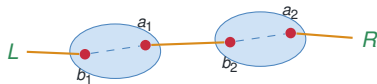
- Relays become free: tightness gives $g = \text{dist} - \rho_u - \rho_v = -2\beta \leq 0$, and $\hat{g} \leq g$.
- So every walk of the search is an L – R path in the graph with edge weights $\max(\hat{g}, 0)$, at no more than the walk's cost:

$$\Delta_{\sigma,H} \geq \Phi := \text{shortest } L\text{--}R \text{ path under } \max(\hat{g}, 0).$$

Φ is only a proof device; it is never computed or reported.

Group the path by true clusters. A cluster is a connected component of the edges with $g \leq 0$ (overlapping total-radius balls, plus blossoms).

- Inside a cluster, $\hat{g} \leq g \leq 0$: moving within a cluster is free in the Φ -graph too.
- So we may reroute to make the path visit each cluster C_j once, entering at b_j and leaving at a_j .
- The landmark bound may add spurious free edges between clusters; the argument never needs them to cost anything.



Innocent side (2/3): the cluster lemma

Cluster lemma

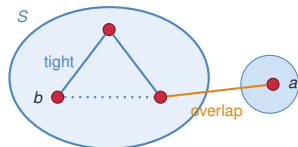
Let $Y(C)$ be the total dual mass of cluster C : every y_v and every y_S inside it, each counted once. For any $a, b \in C$,

$$\text{dist}(b, a) + \rho_a + \rho_b \leq 2 Y(C).$$

Proof sketch. Join b to a by a chain of two kinds of step:

- **between top-level objects:** overlap edges, $\text{dist}(u, v) \leq \rho_u + \rho_v$, $\beta = 0$;
- **inside a blossom:** tight edges of its odd cycle, $\text{dist}(u, v) = \rho_u + \rho_v - 2\beta(u, v)$, recursing into sub-blossoms.

Both kinds satisfy $\text{dist}(u, v) \leq \sum_X y_X$ [the step crosses X], summed over all dual sets X . Because blossoms form a tree of odd cycles, the chain can visit each object in one contiguous stretch. Each X is then crossed at most twice, counting ρ_a and ρ_b as one crossing each. $\square$



A cluster made of a blossom S and an overlapping singleton. The chain from b to a goes around S 's tight cycle, then crosses one overlap edge.

This is where blossom structure is used: sets with $y_S > 0$ must be laminar and carry a tight odd cycle, as sparse blossom guarantees. A small numerical check on dual states with that structure found no violations; for arbitrary LP-optimal duals without it the inequality can fail.

Innocent side (3/3): telescope the L potential

$h_L(x) = \text{dist}(L, x)$ is 1-Lipschitz, $h_L(L) = 0$, and the R end gives $\text{dist}(x, R) \geq \mathcal{D} - h_L(x)$. Along the cluster path:

- **between clusters:** L is a landmark, so $\widehat{\text{dist}} \geq \text{LB} \geq |h_L(a_j) - h_L(b_{j+1})|$; boundary prices are exact;
- **inside clusters:** 1-Lipschitz, then the cluster lemma: $|h_L(a_j) - h_L(b_j)| \leq \text{dist}(b_j, a_j) \leq 2Y(C_j) - \rho_{a_j} - \rho_{b_j}$.

Summing the telescoping differences from 0 to $\mathcal{D}$:

$$\mathcal{D} \leq \sum_{\text{between}} (\text{cost} + \rho_{a_j} + \rho_{b_{j+1}}) + \sum_j (2Y(C_j) - \rho_{a_j} - \rho_{b_j}).$$

The ρ terms cancel ($\rho_L = \rho_R = 0$), and $\sum_j Y(C_j) \leq W^*$ by strong duality.



h_L must climb from 0 to $\mathcal{D}$. Hops pay for the climb outside clusters; clusters contribute at most their dual mass. At most $2W^*$ of the climb is free.

Result

$$\Delta_{\sigma, H} \geq \Phi \geq \mathcal{D} - 2W^* \geq \mathcal{D} - 2W(E).$$

Together with $\Delta_{\sigma, H} \leq \Delta$, the bound keeps the doubled and quadrupled exponents.

What the yoking guarantee needs, and what it predicts

What the proof needs

- L (or R) in the landmark set. fastgap always adds both.
- h_L, h_R not saturated: store as `int64`, or assert $\text{cap} \geq \mathcal{D}$.
- Distances never overestimated. Correlated matching reweights the graph, which would make stale tables break (C).
- An outer decoder that returns a minimum-weight correction: true for 1D and 2D yokes, *not* for BP or for non-matchable 3D yokes.

What it predicts

- Only the L potential matters for the exponent: even `boundary_only` keeps it. Other landmarks buy tightness, i.e. the prefactor.
- Same slope in d as the exact gap; the fitted bases (8, 50) may drop, more in 2D, where a wrong rectangle has 2–3 innocents.

Experiment E2

Per-patch triples (Δ , $\Delta_{\sigma,H}$, fail) from 10 d -round memory windows, then the paper's outer simulation once per estimator. Fit the 1D and 2D bases and check that the slope in d matches.

Summary

- Δ is the soft output applications want; the usual second decode pays for the empty volume between the few small clusters of defects.
- $M^* \oplus M^c$ is one alternating path. The decoder's duals turn its cost into a sum of non-negative **reduced costs**, so one Dijkstra on the defects gives $\Delta_\sigma \leq \Delta$, blossoms included.
- Long hops are priced by landmark lower bounds, so we report $\Delta_{\sigma,H} \leq \Delta_\sigma \leq \Delta$ (not Δ_σ itself) and never touch the empty volume. Re-pricing the walk gives $\Delta_h \geq \Delta$.
- **Measured:** 9–45 $\times$ faster than the class-forced decode; 45 $\times$ at $d = 17$ (12 μ s vs. 536 μ s median).
- **Yoking:** $\Delta_{\sigma,H} \leq \Delta$ and $\Delta_{\sigma,H} \geq \mathcal{D} - 2W^*$, so the bound keeps the 1D/2D distance exponents; only the prefactor can change.

Next

- Latency on a pinned Linux server, and end to end.
- How often $\Delta_{\sigma,H} = \Delta$ vs. table radius and landmarks.
- Applications: cultivation (relative retry rate), yoking (fitted bases and slope vs. exact gap), qLDPC concatenation (soft vs. hard).

Refs: Higgott & Gidney 2025 (sparse blossom); Edmonds 1965; Goldberg & Harrelson 2005 (landmarks); Gidney et al. 2024 (cultivation), 2025 (yoking); Meister et al. 2024.